

Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Project Title

MediLink – Smart Doctor Appointment & Healthcare Management System.

Submitted By

Name: Md. Ekramul Haque

ID: 11220320838

Name: Emam Mahadi Hasan

ID: 11220320832

Name: Lamia Ahmed Zim

ID: 11220320952

Semester: 8A

Submitted To

Md. Riaz Mahmud

Assistant Professor

Department of Computer Science and

Engineering , NUBTK

CSE 4104

Mobile Computing Lab

Team Name: CSE4204-T06

Team Leader: Md.Ekramul Haque

Department of Computer Science and Engineering

Northern University of Business and Technology

Khulna

Khulna 9208,Bangladesh

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Introduction:

Healthcare systems in many hospitals and clinics are still dependent on traditional and manual processes for managing appointments, patient records, prescriptions, and doctor-patient communication. These outdated systems often create serious problems such as long waiting times, double booking of appointments, loss of medical records, poor communication between patients and doctors, and inefficient hospital management. Patients usually need to physically visit hospitals or clinics just to take an appointment, which wastes both time and money. In emergency situations, this process becomes even more difficult and stressful for patients. To solve these problems, this project proposes an AI-powered Smart Doctor Appointment and Healthcare Management System named **MediLink**. The system is designed to provide a complete digital healthcare ecosystem where patients, doctors, and hospital administrators can interact efficiently through a secure online platform. The main purpose of this system is to simplify healthcare services by allowing patients to search for doctors, book appointments in real time, join online video consultations, receive digital prescriptions, and maintain medical records securely. At the same time, doctors will be able to manage their schedules, patient history, prescriptions, and consultation sessions more effectively. Hospital administrators will also gain advanced monitoring and management capabilities through an analytics dashboard. Unlike traditional appointment systems, MediLink will integrate modern technologies such as Artificial Intelligence (AI), real-time communication, cloud storage, digital prescriptions, and secure authentication systems. AI-based symptom checking and smart doctor recommendation features will make the platform more intelligent and user-friendly. The proposed system aims to create a modern, scalable, secure, and efficient healthcare management solution that can significantly improve the quality of healthcare services and patient experience.

Problem Statement:

Traditional healthcare systems often rely on manual appointment booking and paper-based record management. This approach can lead to scheduling conflicts, long waiting times, misplaced records, and inefficient communication between patients and healthcare providers.

Patients frequently face difficulties in finding available doctors and booking appointments conveniently. Healthcare institutions also struggle with managing patient information,

appointment schedules, and medical records efficiently. Therefore, a digital solution is required to streamline healthcare operations and improve service quality.

Objectives

The primary objective of this project is to develop a complete and intelligent healthcare management system that can digitize and automate hospital and clinic services. The project focuses on improving healthcare accessibility, reducing patient waiting time, enhancing communication, and ensuring secure data management. The specific objectives of the project are described below:

1. **To Develop an Online Doctor Appointment System :** The project aims to create an online appointment booking platform where patients can easily search for doctors based on specialization, experience, location, and availability. Patients will be able to book appointments remotely without physically visiting the hospital or clinic.
2. **To Implement Real-Time Appointment Slot Management:** One of the major goals of the project is to prevent double booking and scheduling conflicts. The system will use real-time slot tracking and smart scheduling algorithms to ensure that appointment slots are updated instantly for all users.
3. **To Provide Digital Prescription Facilities :** The project will introduce a secure digital prescription system where doctors can generate prescriptions electronically. Patients will be able to download and access their prescriptions anytime from the platform.
4. **To Integrate Video Consultation Services:** The system will support online video consultation features so that patients can communicate with doctors remotely. This feature will be especially useful for emergency situations, rural patients, and follow-up consultations.
5. **To Implement AI-Based Symptom Checking:** The project aims to integrate Artificial Intelligence features that can analyze patient symptoms and suggest possible diseases or

recommend suitable doctors. This feature will improve healthcare accessibility and support decision-making.

6. **To Ensure Secure Healthcare Data Management:** Patient medical data is highly sensitive and must be protected. Therefore, the system will implement advanced security mechanisms such as JWT authentication, password encryption, role-based access control, and secure cloud storage.
7. **To Improve Hospital Management Efficiency:** The project also aims to help hospitals and clinics manage appointments, payments, schedules, prescriptions, and patient records more efficiently through automation and analytics.

Scope of the Project:

The Smart Doctor Appointment & Healthcare Management System is designed to provide a comprehensive digital platform that improves the efficiency of healthcare services for patients, doctors, and administrators. The system aims to replace traditional paper-based and manual healthcare management processes with a secure, automated, and user-friendly solution.

The scope of this project includes the development of several core modules that support healthcare operations and patient care management. Patients will be able to create accounts, securely log into the system, manage their personal profiles, search for doctors based on specialization, and book appointments according to doctor availability. They will also be able to view appointment histories, access digital prescriptions, and maintain their medical information through a centralized platform.

Doctors will have access to a dedicated dashboard where they can manage their professional profiles, update their schedules, review appointment requests, access patient information, and generate digital prescriptions. The system will help doctors organize their daily activities more efficiently while reducing paperwork and administrative burden.

Administrators will be responsible for managing the overall system. They will be able to monitor users, manage doctor and patient accounts, oversee appointment activities, generate reports, and ensure the smooth operation of the platform. The administrative module will provide complete control over system management and data monitoring.

The system will also include appointment scheduling and management functionalities that allow patients to book, modify, or cancel appointments. Doctors can approve, reject, or reschedule appointments based on their availability. Automated notifications and appointment reminders will be sent to users to reduce missed appointments and improve communication between patients and healthcare providers.

Additionally, the project will support secure authentication and authorization mechanisms to protect sensitive patient and healthcare data. Medical records and prescription information will be stored in a centralized database to ensure data consistency, accessibility, and reliability.

This project primarily focuses on appointment management, patient record management, doctor management, prescription generation, and administrative control. However, advanced healthcare services such as emergency response systems, ambulance tracking, AI-based disease diagnosis, telemedicine video consultations, health insurance integration, and wearable device connectivity are beyond the scope of the current version and may be considered for future development.

By implementing these functionalities, the system aims to improve healthcare service quality, reduce operational inefficiencies, enhance patient satisfaction, and support the digital transformation of healthcare institutions.